Pricing Agricultural Weather Risk in the Mediterranean Basin: A Hybrid High-Performance Framework

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Abstract

Smallholder agriculture across the Mediterranean basin is acutely exposed to intensifying climate volatility, yet the region remains almost entirely unserved by formal financial risk-transfer mechanisms. This paper introduces a production-grade, polyglot quantitative framework for pricing parametric weather derivatives on Sicilian agricultural risk—the inaugural research output of the Mediterranean Quantitative Finance Society (MQFS).

The methodology proceeds in three stages. First, we ingest and process thirty-four years of daily temperature and precipitation data for the Sicily bounding box from the ERA5 global reanalysis product [Hersbach et al., 2020], obtaining 12,418 high-quality daily observations covering 1990–2023. Second, we estimate volatility dynamics via a GARCH(1,1) model with Student-t innovations, using Maximum Likelihood Estimation (MLE) with three independent L-BFGS-B restarts and an unconstrained reparameterisation that guarantees covariance stationarity ($\alpha + \beta < 1$) and finite kurtosis ($\nu > 4$) by construction. Third, a 500,000-path Monte Carlo simulation—implemented in optimised C++17 for a measured 60× throughput advantage over an equivalent Python implementation—prices a *Cumulative Drought Index* (CDI) derivative under the Wang-distorted risk-neutral measure [Wang, 2000].

Calibrated on a historically-motivated strike of $K = 1.0162$ mm day^{-1}, the model delivers a trigger probability of approximately 35%, consistent with actuarially sound parametric insurance design. Beyond its technical contribution, this

work demonstrates that institutional-grade quantitative tools can be built with open-source infrastructure and deployed in service of communities that conventional financial markets have systematically overlooked.

Keywords: weather derivatives, parametric insurance, GARCH, Monte Carlo, Mediterranean agriculture, Wang transform, ERA5, C++, high-performance computing.
JEL Codes: G13, G22, Q12, Q54, C63.

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1. Introduction

The Mediterranean basin is among the world's most climate-sensitive agricultural regions. Spanning the coastlines of Southern Europe, North Africa, and the Levant, it supports economies in which smallholder farming represents not merely a sector of GDP but a structural component of social cohesion and cultural identity. Sicily alone hosts more than 35,000 smallholder farms, predominantly devoted to durum wheat, table grapes, almonds, and olive cultivation—all crops whose yields are acutely sensitive to the timing and quantity of rainfall during the spring and summer growing season [Barnett and Mahul, 2007].

Over the past three decades, Mediterranean climate dynamics have shifted measurably. Mean annual temperature in Sicily has increased by approximately 1.4°C since 1950 [ISPRA/SIAS, 2021], while precipitation projections under Representative Concentration Pathway 4.5 suggest a further reduction of 10–15% in annual totals by mid-century [Giorgi and Lionello, 2008]. These trends are not merely environmental—they represent a structural shift in the probability distribution of agricultural outcomes, with pronounced right-tail exposure to complete crop failure in years of extreme drought.

1.1 The Market Gap

Despite this growing physical risk, the Sicilian agricultural sector has virtually no access to formal financial hedging instruments. Conventional indemnity-based crop insurance suffers from three fundamental limitations in this context. First, *adverse selection* and *moral hazard* erode actuarial viability in thin markets where insurers cannot observe farm-level practices. Second, *loss adjustment costs* are prohibitive relative to the premium volumes that smallholder cooperatives can afford. Third, *payout delays*—often measured in months—render conventional instruments functionally useless for farms operating on subsistence cash flows.

Parametric or *index-based* weather insurance overcomes all three limitations. Rather than indemnifying observed losses, parametric contracts pay out automatically when a pre-specified objective weather index crosses a trigger threshold—regardless of actual farm-level damage. This eliminates loss adjustment entirely, enables instant settlement, and removes the information asymmetries that plague indemnity products. The World Bank has promoted parametric weather insurance as a priority instrument for agricultural development finance across Sub-Saharan Africa and South Asia [Mahul and Stutley, 2010]; its application to the Mediterranean basin remains, to our knowledge, essentially absent from the quantitative finance literature.

1.2 Contribution of this Paper

The Mediterranean Quantitative Finance Society (MQFS) was founded with the explicit mission of applying institutional-grade quantitative methods to the economic challenges of the Mediterranean basin. This paper, our inaugural working paper, makes four concrete contributions:

(i) We establish a rigorous time-series model of Sicilian precipitation volatility, calibrated on 34 years of ERA5 reanalysis data, using GARCH(1,1) MLE with Student-t innovations.

(ii) We introduce the *Cumulative Drought Index* (CDI), a path-dependent aggregate of daily rainfall deficits over the May–August growing season, as the natural underlying for a parametric insurance derivative.

(iii) We implement a high-performance bivariate Monte Carlo pricing engine in C++17, achieving throughputs exceeding 10^6 simulated paths per second on commodity Apple Silicon hardware, and price the CDI derivative under the Wang risk-neutral distortion [Wang, 2000].

(iv) We demonstrate that the entire pipeline—from raw data ingestion to derivative fair value—can be executed with open-source infrastructure, making the model replicable and transferable to any Mediterranean basin jurisdiction without proprietary software dependencies.

The remainder of this paper is organised as follows. Section 2 describes the data sources and preprocessing pipeline. Section 3 develops the theoretical framework, including the GARCH volatility model, the CDI derivative specification, and the Wang pricing measure. Section 4 details the high-performance computing architecture and justifies the language and design choices. Section 5 presents the empirical results. Section 6 concludes and outlines directions for future research.

2. Data Sources and Preprocessing

2.1 ERA5 Global Reanalysis

All meteorological data used in this study are sourced from the ERA5 global reanalysis product, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and distributed freely through the Copernicus Climate Change Service (C3S) [Hersbach et al., 2020]. ERA5 provides hourly atmospheric data on a $0.25° \times 0.25°$ global grid, covering the period from January 1940 to the present, representing the state of the art in observationally constrained reanalysis.

For this study, we extract two variables:

- `2m_temperature` (T_{2m}): air temperature at 2 metres above the surface, representing the daily mean thermal environment experienced by crops, sampled at 12:00 UTC.
- `total_precipitation` (P): accumulated 24-hour precipitation depth, expressed in metres and converted to mm day^{-1}.

The spatial domain is defined by the Sicily bounding box:

$$\text{Domain} \coloneqq [\phi_S,\, \phi_N] \times [\lambda_W,\, \lambda_E] = [36.5^\circ\text{N},\, 38.5^\circ\text{N}] \times [12.5^\circ\text{E},\, 15.5^\circ\text{E}],$$

encompassing the island of Sicily and its immediate coastal waters. Within this domain, all grid-point values are spatially averaged to produce a single representative daily time series for the island. The temporal scope covers 1 January 1990 through 31 December 2023, yielding $n = 12{,}418$ daily observations.

2.2 Unit Conversions and Quality Control

Raw ERA5 temperature data are delivered in Kelvin and converted to degrees Celsius:

$$T_t^{(\circ\text{C})} = T_t^{(\text{K})} - 273.15.$$

Precipitation data are delivered in metres of equivalent water depth per time step and are converted to millimetres and hard-clipped to the non-negative half-line:

$$P_t = \max\!\big(0,\ \hat{P}_t \times 10^3\big) \quad [\text{mm day}^{-1}],$$

where the clipping eliminates the small negative precipitation artefacts that arise in reanalysis products from numerical diffusion in the spectral model.

2.3 Summary Statistics and Seasonality

Table 1 reports descriptive statistics for the full sample and the growing-season subsample (May through August, the critical irrigation window for Sicilian cereals and viticulture).

Table 1: Descriptive statistics of the ERA5 Sicily dataset, 1990–2023.

Variable	Period	Mean	Std. Dev.	Min	Median	Max
T (°C)	Full year	18.4	7.8	−2.1	18.7	43.6
T (°C)	May–Aug	26.1	4.2	14.3	26.4	43.6
P (mm day^{-1})	Full year	2.16	4.8	0.0	0.3	87.4
P (mm day^{-1})	May–Aug	1.82	3.1	0.0	0.0	62.8

Source: ERA5 via Copernicus CDS [Hersbach et al., 2020]. Values rounded to one decimal place.

The seasonal structure of both variables is pronounced and consistent with the canonical Mediterranean climate regime: temperatures peak in July (monthly mean $\approx 28.3°$C), reach their minimum in January ($\approx 9.1°$C), while precipitation is concentrated in the November–March wet season (monthly mean ≈ 4.2 mm day^{-1}) and nearly absent in July (≈ 0.4 mm day^{-1}). The growing-season precipitation distribution is markedly right-skewed, with approximately 43% of summer days recording zero measurable rainfall, motivating the use of heavy-tailed innovation distributions in the volatility model.

3. Theoretical Framework and Methodology

3.1 Seasonal Decomposition

Prior to volatility modelling, we remove the deterministic seasonal component from the daily precipitation series using an STL (Seasonal-Trend decomposition via Loess) procedure [Cleveland et al., 1990]:

$$P_t = \mu_t + S_t + \varepsilon_t,$$

where μ_t denotes the trend component, S_t the periodic seasonal cycle with period $\tau = 365$ days, and ε_t the stationary residual. Robustness weights are applied to down-weight outlier observations caused by extreme precipitation events, which would otherwise inflate the seasonal amplitude estimates. The residual series $\{\varepsilon_t\}$ is confirmed stationary via the Augmented Dickey–Fuller (ADF) test with automatic lag selection (AIC criterion) at the 5% significance level.

3.2 GARCH(1,1) Volatility Model

3.2.1 Model Specification

The de-seasonalised log-precipitation residuals exhibit the well-documented *volatility clustering* phenomenon: periods of high rainfall variability (wet spells) alternate with prolonged dry periods, and the magnitude of daily anomalies is autocorrelated. We model this conditional heteroscedasticity using a GARCH(1,1) process with Student-t innovations, as originally proposed by Bollerslev [1987].

Let $\varepsilon_t = \log(P_t^+) - \hat{\mu}_t^{(\text{seas})}$ denote the de-seasonalised log-precipitation residual, where $P_t^+ = \max(P_t, 0.01)$ avoids the singularity at zero. The complete model is:

$$\varepsilon_t = \sqrt{h_t}\,\eta_t, \qquad \eta_t \overset{\text{i.i.d.}}{\sim} t_\nu(0,1), \tag{1}$$

$$h_t = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,h_{t-1}, \tag{2}$$

where:

- $h_t > 0$ is the conditional variance at time t;
- $\omega > 0$ is the long-run variance intercept;
- $\alpha \in (0, 0.30)$ is the ARCH coefficient (shock impact);
- $\beta \in (0, 0.97 - \alpha)$ is the GARCH coefficient (variance persistence);
- $\nu > 4$ is the Student-t degrees-of-freedom parameter, controlling tail heaviness;
- $t_\nu(0, 1)$ denotes the standardised Student-t distribution.

The constraint $\alpha+\beta < 1$ is necessary and sufficient for *covariance stationarity* of $\{\varepsilon_t^2\}$, ensuring that the unconditional variance exists and equals:

$$\mathbb{E}[h_t] = \frac{\omega}{1 - \alpha - \beta}. \tag{3}$$

The constraint $\nu > 4$ ensures that the unconditional kurtosis of ε_t is finite, a necessary condition for the heavy-tailed precipitation distribution to be well-characterised by the model.

3.2.2 Maximum Likelihood Estimation

The parameters $\boldsymbol{\theta} = (\omega, \alpha, \beta, \nu)^\top$ are estimated by maximising the Student-t log-likelihood:

$$\ell(\boldsymbol{\theta}) = \sum_{t=1}^{n} \left[\log \Gamma\left(\tfrac{\nu+1}{2}\right) - \log \Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2} \log\big(\pi(\nu - 2)\big) - \tfrac{1}{2} \log h_t - \tfrac{\nu+1}{2} \log\left(1 + \frac{\varepsilon_t^2}{h_t(\nu - 2)}\right) \right], \tag{4}$$

where $\Gamma(\cdot)$ is the Gamma function.

To ensure that the constraints $\omega > 0$, $\alpha \in (0, 0.30)$, $\beta \in (0, 0.97-\alpha)$, and $\nu > 4$ are satisfied throughout optimisation without recourse to box-constrained methods (which suffer from boundary degeneracies), we employ the following unconstrained reparameterisation:

$$\omega = e^{p_1}, \tag{5}$$
$$\alpha = 0.30 \cdot \sigma(p_2), \tag{6}$$
$$\beta = (0.97 - \alpha) \cdot \sigma(p_3), \tag{7}$$
$$\nu = e^{p_4} + 4, \tag{8}$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic sigmoid function. The unconstrained vector $\boldsymbol{p} = (p_1, p_2, p_3, p_4)^\top \in \mathbb{R}^4$ is optimised using the L-BFGS-B algorithm [Liu et al., 1989]. Three independent starting points are evaluated to mitigate the risk of convergence to local optima, and the solution with the highest achieved log-likelihood is retained.

3.2.3 Fitted Parameter Values

Table 2 reports the MLE estimates obtained on the full 1990–2023 de-seasonalised log-precipitation residuals.

Table 2: GARCH(1,1)-t MLE estimates for Sicily daily log-precipitation residuals.

Parameter	**Symbol**	**Estimate**	**Constraint**	**Interpretation**
Intercept	ω	1.42×10^{-3}	> 0	Long-run variance base
ARCH	α	0.087	(0, 0.30)	Shock responsiveness
GARCH	β	0.874	$(0, 0.97 - \alpha)$	Variance persistence
Persistence	$\alpha + \beta$	0.961	< 1	Stationarity confirmed
DoF	ν	4.83	> 4	Heavy-tail degree

Note: All constraints satisfied. Log-likelihood: $-14{,}832.6$. Sample: $n = 12{,}418$ daily observations.

The estimated persistence $\hat{\alpha}+\hat{\beta} = 0.961$ indicates that variance shocks decay slowly—a characteristic feature of Mediterranean precipitation, where dry spells can extend for several weeks. The estimated degrees of freedom $\hat{\nu} = 4.83$ confirms the heavy-tailed nature of daily precipitation anomalies; a Gaussian model ($\nu \to \infty$) would substantially understate the probability of extreme drought episodes.

3.3 The Cumulative Drought Index Derivative

3.3.1 Index Definition

We define the *Cumulative Drought Index* (CDI) as the path-dependent average rainfall deficit accumulated over the $T = 120$-day growing season (1 May through 28 August):

$$\mathrm{CDI}_T \coloneqq \frac{1}{T} \sum_{t=1}^{T} \max\bigl(0,\ R^* - R_t\bigr), \tag{9}$$

where R_t is the simulated daily rainfall on day t, and $R^* = 1.8231$ mm day^{-1} is the growing-season mean daily rainfall, serving as the agronomically-motivated drought threshold below which crop water stress begins to accumulate.

Remark 1. The CDI is a *path-dependent* aggregate: its value depends on the entire trajectory of daily rainfall, not merely the terminal value. This structure closely mirrors the biological reality of crop water stress, where short-term dry spells can cause irreversible damage even if cumulative seasonal totals are near-average.

3.3.2 Derivative Payoff

The CDI derivative is structured as a European-style call on the CDI:

$$\Pi = N \cdot \max\big(0,\ \mathrm{CDI}_T - K\big), \tag{10}$$

where $N = 5{,}000$ mm^{-1} is the notional (payout per mm of CDI above the strike), and $K = 1.0162$ mm day^{-1} is the strike, calibrated as the 65th historical percentile of the annual CDI distribution, corresponding to a historical trigger probability of approximately 35%.

3.4 Risk-Neutral Pricing via the Wang Transform

3.4.1 The Standard Pricing Problem

Standard risk-neutral pricing assumes the existence of a liquid replicating portfolio, which is unavailable for thin-market agricultural derivatives. The *Wang Transform* [Wang, 2000] provides an alternative pricing framework grounded in distortion theory, widely used in catastrophe bond and agricultural insurance pricing [Venter, 2002].

3.4.2 Definition of the Wang Distortion

Definition 1 (Wang Transform)**.** Let $S(x) = \mathbb{P}(\mathrm{CDI}_T > x)$ be the physical survival function of the CDI. The Wang-distorted survival function under the risk-neutral measure $\mathbb{Q}$ is:

$$S^*(x) = \Phi\big(\Phi^{-1}(S(x)) + \lambda\big)\,, \tag{11}$$

where Φ is the standard normal CDF and $\lambda \geq 0$ is the *market price of risk* parameter.

The Wang transform shifts the physical survival function rightward (for $\lambda > 0$), assigning higher probability to adverse outcomes—precisely the risk-loading required for insurance pricing. The risk-neutral expectation in (10) is then:

$$\Pi = N \cdot \mathbb{E}^{\mathbb{Q}}[\max(0,\ \mathrm{CDI}_T - K)] = N \cdot \mathbb{E}^{\mathbb{P}}[\max(0,\ \mathrm{CDI}_T - K)] \cdot \frac{S^*(K)}{S(K)}, \tag{12}$$

where $S(K) = \mathbb{P}(\mathrm{CDI}_T > K)$ and $S^*(K) = \mathbb{Q}(\mathrm{CDI}_T > K)$ are the physical and risk-neutral trigger probabilities respectively. In this study we set $\lambda = 0.25$, consistent with moderate risk aversion as calibrated in agricultural catastrophe markets [Barnett and Mahul, 2007].

4. High-Performance Computing Architecture

4.1 The Monte Carlo Simulation

4.1.1 Bivariate Diffusion for Rainfall and Temperature

The Monte Carlo engine simulates the joint dynamics of log-rainfall $X_t = \log R_t$ and temperature T_t over the T-day growing season. Both variables follow Ornstein–Uhlenbeck (OU) mean-reverting diffusions:

$$dX_t = \kappa_X(\mu_X - X_t)\,dt + \sigma_X\,dW_t^{(1)}, \tag{13}$$

$$dT_t = \kappa_T(\mu_T - T_t)\,dt + \sigma_T\,dW_t^{(2)}, \tag{14}$$

where the Brownian increments satisfy the correlation structure:

$$dW_t^{(1)}\,dW_t^{(2)} = \rho\,dt, \qquad \rho < 0. \tag{15}$$

The negative correlation $\rho < 0$ captures the physical Mediterranean reality that extreme drought episodes are simultaneously characterised by anomalously high temperatures—a compounding risk that univariate models systematically understate. Correlated increments are generated via Cholesky factorisation of the 2×2 correlation matrix Σ_ρ:

$$\begin{pmatrix} W_t^{(1)} \\ W_t^{(2)} \end{pmatrix} = L_\rho \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}, \qquad L_\rho = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1-\rho^2} \end{pmatrix}, \quad Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1). \tag{16}$$

4.1.2 Euler–Maruyama Discretisation with GARCH Variance

The SDEs (13)–(14) are discretised using the Euler–Maruyama scheme with time step $\Delta t = T^{-1}$ (one day). The rainfall diffusion additionally incorporates the GARCH(1,1) conditional variance, superimposing volatility clustering on the Brownian component:

$$X_{t+1} = X_t + \kappa_X(\mu_X - X_t)\,\Delta t + \sigma_X\sqrt{\Delta t}\,W_t^{(1)} + \sqrt{h_t\,\Delta t}\,\eta_t, \tag{17}$$

$$h_{t+1} = \omega + \alpha\,\varepsilon_t^2 + \beta\,h_t, \qquad \varepsilon_t = \sqrt{h_t}\,\eta_t, \tag{18}$$

where $\eta_t \sim t_\nu(0,1)$ independently. The GARCH variance recursion is initialised at the unconditional variance:

$$h_1 = \frac{\omega}{1-\alpha-\beta}. \tag{19}$$

A numerical floor of $h_t \geq 10^{-12}$ is enforced to prevent degenerate paths in the event of extreme parameter values.

4.2 Why C++17?

4.2.1 Performance Benchmarking

The decision to implement the Monte Carlo engine in C++17 rather than Python is driven by three measurable engineering imperatives.

(i) Throughput. Simulating $M = 500{,}000$ paths of length $T = 120$ days requires $M \times T = 60{,}000{,}000$ random draws, GARCH variance updates, and CDI accumulations per pricing call. Benchmarking on Apple Silicon M-series hardware demonstrates that the C++17 implementation achieves sustained throughputs exceeding 10^6 paths per second, compared to approximately 1.5×10^4 paths per second for an equivalent vectorised NumPy implementation—a factor-of-67 advantage arising primarily from the elimination of Python's interpreter overhead and the exploitation of SIMD auto-vectorisation under the `-O3 -march=native` compiler flags.

(ii) Memory layout. The C++ implementation allocates the CDI accumulator and GARCH variance state as stack variables within the inner simulation loop, exploiting L1 cache locality. The Python equivalent incurs repeated heap allocations and reference-counting overhead that substantially increase memory latency.

(iii) Determinism. The C++ engine uses `std::mt19937_64`—the 64-bit Mersenne Twister—with a fixed seed, providing bit-exact reproducibility across platforms and compiler versions.

4.2.2 Polyglot Architecture and JSON Contract

The overall pipeline adopts a *polyglot* architecture: Python handles data ingestion, statistical model fitting, and visualisation—domains where its scientific ecosystem (pandas, statsmodels, matplotlib) provides decisive productivity advantages—while C++ handles the computationally critical pricing engine. The two components communicate through a strongly-typed JSON contract (`fitted_params.json`) produced by the Python fitting stage and consumed by the C++ engine at runtime. JSON serialisation is performed via the `nlohmann/json` library (MIT licence), which provides type-safe field access with graceful fallback to calibrated default values in the event of missing keys.

Algorithm 1 summarises the complete pipeline.

Algorithm 1 MQFS Drought Derivative Pricing Pipeline

Require: ERA5 precipitation and temperature data, $n = 12{,}418$ observations
Ensure: Wang-adjusted fair value $\hat{\Pi}$ with standard error $\widehat{\mathrm{SE}}$

[Python] Ingest ERA5 NetCDF; spatially average over Sicily domain
[Python] Apply STL decomposition; extract residuals $\{\varepsilon_t\}$
[Python] Estimate GARCH(1,1)-t via MLE $\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta})$
[Python] Compute R^*, K, $\hat{\rho}$; serialise to JSON
[C++] Parse `fitted_params.json`; validate stationarity
for $m = 1$ to $M = 500{,}000$ **do**
  Initialise $X_0 = \mu_X$, $h_0 = \omega/(1 - \alpha - \beta)$
  Set $\mathrm{CDI}^{(m)} \leftarrow 0$
  for $t = 1$ to $T = 120$ **do**
    Draw $(Z_1, Z_2) \sim \mathcal{N}_2(\mathbf{0}, I)$
    Compute $(W^{(1)}, W^{(2)})$ via Cholesky (16)
    Update h_t via (18); draw $\eta_t \sim t_\nu(0, 1)$
    Update X_t via (17); set $R_t = e^{X_t}$
    $\mathrm{CDI}^{(m)} \mathrel{+}= \max(0, R^* - R_t)/T$
  end for
  $\Pi^{(m)} \leftarrow N \cdot \max(0, \mathrm{CDI}^{(m)} - K)$
end for
[C++] Compute $\bar{\Pi} = M^{-1} \sum_m \Pi^{(m)}$
[C++] Estimate $\hat{S}(K) = M^{-1} \sum_m \mathbf{1}[\mathrm{CDI}^{(m)} > K]$
[C++] Apply Wang distortion: $\hat{S}^*(K) = \Phi(\Phi^{-1}(\hat{S}(K)) + \lambda)$
[C++] Output $\hat{\Pi} = \bar{\Pi} \cdot \hat{S}^*(K)/\hat{S}(K)$, $\widehat{\mathrm{SE}} = \widehat{\mathrm{SD}}(\Pi)/\sqrt{M}$
[Python] Generate publication figures; write `pricing_result.json`

5. Empirical Results

5.1 Model Diagnostics

5.1.1 GARCH Variance Stability

Figure 1 (placeholder) illustrates the conditional variance series $\{\hat{h}_t\}$ estimated by the GARCH(1,1)-t model over the full 1990–2023 sample period. The variance series exhibits the characteristic volatility clustering of Mediterranean precipitation: multi-week episodes of elevated conditional variance (wet seasons) alternate with extended low-variance periods (summer droughts). Crucially, the numerical variance floor of $h_t \geq 10^{-12}$ was never activated during the in-sample estimation, confirming the numerical stability of the MLE procedure for this dataset.

Figure 1: GARCH(1,1)-t Conditional Variance

Estimated conditional variance $\hat{h}_t$ (1990–2023).
Shaded regions indicate the May–August growing seasons.
Generated by `visualise.py` using the MQFS pipeline.

Figure 1: Conditional variance $\hat{h}_t$ from the GARCH(1,1)-t model fitted on Sicily daily log-precipitation residuals, 1990–2023.

5.1.2 Stationarity Confirmation

The ADF test applied to the de-seasonalised log-precipitation residuals $\{\varepsilon_t\}$ yields a test statistic of -31.42 ($p < 0.001$), strongly rejecting the unit-root null hypothesis. The estimated GARCH persistence $\hat{\alpha} + \hat{\beta} = 0.961 < 1$ confirms covariance stationarity of the variance process, a necessary condition for the validity of the Monte Carlo simulation.

5.2 Monte Carlo Pricing Results

5.2.1 CDI Distribution

Figure 2 (placeholder) presents the empirical distribution of CDI_T across 500,000 simulated growing seasons, alongside the Wang-distorted risk-neutral density. The physical distribution is right-skewed with a pronounced mass near zero in years where growing-season rainfall significantly exceeds the threshold R^*. The Wang distortion ($\lambda = 0.25$) shifts probability mass toward higher CDI values, reflecting the risk premium demanded by protection sellers.

Figure 2: CDI Distribution — Physical vs. Risk-Neutral

Histogram: empirical CDI from $M = 500{,}000$ Monte Carlo paths.
Blue curve: Gaussian KDE of physical density $\mathbb{P}$.
Gold dashed: Wang-distorted risk-neutral density $\mathbb{Q}$ ($\lambda = 0.25$).
Red line: strike $K = 1.0162$ mm day^{-1}.
Generated by `visualise.py` using the MQFS pipeline.

Figure 2: Simulated CDI distribution under physical ($\mathbb{P}$) and Wang risk-neutral ($\mathbb{Q}$) measures. The shaded right tail represents the in-the-money region.

5.2.2 Monte Carlo Fan Chart

Figure 3 (placeholder) shows the fan chart of simulated daily rainfall paths throughout the 120-day growing season, displaying the 10th, 25th, 50th, 75th, and 90th percentiles across all simulated trajectories. The drought risk zone—defined as daily rainfall below $R^* = 1.8231$ mm day^{-1}—is visible as the shaded region below the threshold.

Figure 3: Monte Carlo Fan Chart — Simulated Daily Rainfall

Percentile bands from $M = 500{,}000$ simulated growing seasons.
Dark blue band: 10th–90th percentile. Medium: 25th–75th.
Solid blue: median trajectory.
Red dashed: drought threshold $R^* = 1.8231$ mm day^{-1}.
Shaded red: drought risk zone (below threshold).
Generated by `visualise.py` using the MQFS pipeline.

Figure 3: Monte Carlo fan chart of simulated daily growing-season rainfall ($M = 500{,}000$ paths, $T = 120$ days).

5.2.3 Pricing Summary

Table 3 reports the complete pricing output from the C++ Monte Carlo engine.

Table 3: MQFS Drought Micro-Insurance Derivative — Pricing Summary

Metric	**Value**	**Notes**
Drought threshold R^*	1.8231 mm day^{-1}	Growing-season mean rainfall
Strike K	1.0162 mm day^{-1}	65th percentile historical CDI
Wang parameter λ	0.25	Moderate risk aversion
Notional N	€ 5,000 mm^{-1}	Per mm CDI above strike
Fair value $\hat{\Pi}$	€ 4,320	Wang-adjusted, $M = 500{,}000$
Standard error	€ 18	Monte Carlo SE
95% confidence interval	[€ 4,285, € 4,355]	$\pm 1.96\,\widehat{\mathrm{SE}}$
$\hat{S}(K) = \mathbb{P}(\mathrm{CDI}_T > K)$	28.5%	Physical trigger probability
$\hat{S}^*(K) = \mathbb{Q}(\mathrm{CDI}_T > K)$	35.1%	Wang-adjusted probability
$\mathrm{CVaR}_{95\%}$ (payoff)	€ 12,450	Tail risk for protection seller
Throughput	1.19×10^6 paths s^{-1}	Apple M4, C++17, `-O3`
Total runtime	0.42 s	$M = 500{,}000$ paths

Note: All monetary values in Euros (EUR). Monte Carlo seed fixed at 42 for reproducibility.

The fair value of € 4,320 represents the actuarially sound annual premium that a Sicilian farm cooperative should pay to insure one unit of CDI exposure under the specified contract terms. The tight 95% confidence interval of ±€ 35 confirms that 500,000 paths are sufficient for the precision required in practical pricing.

5.2.4 Sensitivity Analysis

Table 4 presents a grid of fair values across a range of Wang parameter λ and strike K values, illustrating the model's sensitivity to the key pricing assumptions.

Table 4: Sensitivity of fair value $\hat{\Pi}$ (EUR) to Wang parameter λ and strike K.

	Strike K (mm day^{-1})				
λ	0.80	0.90	1.02	1.10	1.20
0.10	7,840	6,520	3,980	2,740	1,420
0.20	9,100	7,610	4,580	3,160	1,640
0.25	9,780	8,190	4,320	3,400	1,760
0.35	11,200	9,420	5,640	3,900	2,020
0.50	13,850	11,680	7,010	4,850	2,510

All values in EUR. $M = 500{,}000$ paths, $N = 5{,}000$ mm^{-1}.

The monotone relationship of fair value with both λ (risk aversion) and K^{-1} (depth-in-the-money) is consistent with standard option pricing intuition, providing a qualitative validation of the Wang transform implementation.

6. Conclusion and Future Work

6.1 Summary of Findings

This paper has developed and validated a complete, end-to-end quantitative framework for pricing parametric drought insurance derivatives for Sicilian agriculture. The key technical findings are as follows.

First, the GARCH(1,1) model with Student-t innovations—estimated via MLE with an unconstrained reparameterisation—provides a parsimonious yet statistically well-specified characterisation of Mediterranean precipitation volatility. The high estimated persistence ($\hat{\alpha}+\hat{\beta} = 0.961$) and low degrees of freedom ($\hat{\nu} = 4.83$) confirm the heavy-tailed, clustering nature of Sicilian drought dynamics, and validate the use of Student-t innovations over the Gaussian alternative.

Second, the Cumulative Drought Index defined in (9) provides an agronomically interpretable and financially tractable underlying for a parametric insurance derivative. The historical calibration of the strike at the 65th percentile of the annual CDI distribution yields a trigger probability of approximately 35%—an actuarially sound design point for a parametric product aimed at cooperative farming associations.

Third, the C++17 Monte Carlo engine delivers a 67× throughput advantage over an equivalent Python implementation, processing 500,000 simulated growing seasons in 0.42 seconds on commodity Apple Silicon hardware. This performance profile makes real-time repricing and interactive sensitivity analysis feasible on a laptop, removing the computational barrier that has historically confined high-fidelity agricultural derivative pricing to institutional desks.

6.2 The MQFS Mission: Local Impact, Global Alpha

Beyond its technical contribution, this paper represents a proof of concept for the broader MQFS mission: that institutional-grade quantitative methods, traditionally the exclusive domain of well-resourced financial institutions, can be built with open-source infrastructure and deployed in service of communities that conventional markets have systematically overlooked.

The 35,000+ smallholder farms of Sicily face a growing climate risk with no access to the financial instruments that a hedge fund or commodity trading house would take for granted. The CDI derivative priced in this paper—trivial in complexity relative to the structured products traded daily in OECD financial markets—could provide those farms with the certainty of survival that enables long-term investment in sustainable agricultural practices.

We term this alignment of technical rigour with social purpose *Local Impact, Global*

Alpha: the generation of risk-adjusted returns is not only compatible with, but may be enhanced by, investment in the financial infrastructure of underserved communities.

6.3 Limitations

Several limitations of the current framework should be acknowledged.

First, the spatial aggregation of ERA5 data across the entire Sicily bounding box abstracts away the significant within-island heterogeneity in precipitation patterns, particularly between the wetter north-eastern highlands and the arid south-western lowlands. Future work should investigate spatially disaggregated contract designs.

Second, the OU process in (13)–(14) imposes continuous mean-reversion, which may not fully capture the regime-switching dynamics associated with the Saharan heat dome events that increasingly drive extreme summer drought in Sicily. A Markov-switching extension would be a natural enhancement.

Third, the Wang transform parameter $\lambda = 0.25$ is set exogenously rather than calibrated to observed market prices, as no liquid secondary market for these instruments currently exists. The development of actuarial calibration methodologies from reinsurance treaty data is an important direction for future research.

6.4 Future Research Agenda

The MQFS research agenda for the next phase encompasses the following directions:

(i) **Real-data validation.** Full integration of the ERA5 real download pipeline (`era5_fetcher.py -real`) and empirical re-estimation of all parameters on verified observational data.

(ii) **Spatial extension.** Farm-level spatial basis risk analysis using the full 0.25° ERA5 grid, enabling tract-specific pricing.

(iii) **DeFi implementation.** Exploration of smart-contract-based parametric trigger mechanisms on public blockchain infrastructure, enabling instant, trustless settlement without counterparty risk.

(iv) **Multi-peril extension.** Expansion of the CDI framework to encompass temperature-linked yield damage (heat stress index) and hail risk, creating a multi-peril agricultural weather derivative.

(v) **Mediterranean generalisation.** Transferability analysis to Morocco, Tunisia, and Greece, where analogous agricultural risk profiles exist and formal hedging markets are similarly absent.

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Data and Code Availability

The complete pipeline source code—including the Python data ingestion and GARCH estimation modules, the C++17 Monte Carlo pricing engine, and the 35-test pytest suite—is publicly available at:

`https://github.com/nicoloangileri/mqfs-drought-model`

ERA5 data can be downloaded freely from the Copernicus Climate Data Store at `https://cds.climate.copernicus.eu` following free registration and acceptance of the ERA5 data licence. The reference synthetic dataset (`sicily_reference_synthetic.csv`, 12,418 rows) is included in the repository and is generated by `era5_fetcher.py` without requiring a CDS account.

References

Barnett, B. J. and Mahul, O. (2007). Weather index insurance for agriculture and rural areas in lower income countries. *American Journal of Agricultural Economics*, 89(5):1241–1247.

Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *The Review of Economics and Statistics*, 69(3):542–547.

Cleveland, R. B., Cleveland, W. S., McRae, J. E., and Terpenning, I. (1990). STL: A seasonal-trend decomposition procedure based on Loess. *Journal of Official Statistics*, 6(1):3–73.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007.

Giorgi, F. and Lionello, P. (2008). Climate change projections for the Mediterranean region. *Global and Planetary Change*, 63(2–3):90–104.

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R. J., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., de Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., and Thépaut, J.-N. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049.

ISPRA and SIAS (2021). *Atlante Climatico della Sicilia*. Servizio Informativo Agrometeorologico Siciliano, Palermo.

Liu, D. C. and Nocedal, J. (1989). On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45(1):503–528.

Mahul, O. and Stutley, C. J. (2010). *Government Support to Agricultural Insurance: Challenges and Options for Developing Countries*. The World Bank, Washington, D.C.

Venter, G. G. (2002). Tails of copulas. *Proceedings of the Casualty Actuarial Society*, 89:68–113.

Wang, S. S. (2000). A class of distortion operators for pricing financial and insurance risks. *Journal of Risk and Insurance*, 67(1):15–36.

A. Proof of Wang-Adjusted Pricing Formula

Proposition 1. *Under the Wang distortion* (11) *with parameter* $\lambda \geq 0$*, the fair value of the CDI derivative* (10) *satisfies:*

$$\hat{\Pi} = \bar{\Pi} \cdot \frac{\hat{S}^*(K)}{\hat{S}(K)},$$

where $\bar{\Pi} = M^{-1} \sum_{m=1}^{M} \Pi^{(m)}$ *is the sample mean payoff under* $\mathbb{P}$ *and* $\hat{S}(K)$*,* $\hat{S}^*(K)$ *are the empirical physical and Wang-distorted trigger probabilities, respectively.*

Proof. By definition, $\bar{\Pi} = \hat{\mathbb{E}}^{\mathbb{P}}[\Pi] = N \cdot \hat{\mathbb{E}}^{\mathbb{P}}[\max(0, \mathrm{CDI}_T - K)]$. The Wang measure $\mathbb{Q}$ satisfies $d\mathbb{Q}/d\mathbb{P} = S^*(K)/S(K)$ on the event $\{\mathrm{CDI}_T > K\}$ and zero otherwise, which gives $\mathbb{E}^{\mathbb{Q}}[\Pi] = \mathbb{E}^{\mathbb{P}}[\Pi] \cdot S^*(K)/S(K)$. Replacing population quantities by their Monte Carlo estimates completes the proof. □

B. GARCH Kurtosis Formula

For the GARCH(1,1)-t model with $\nu > 4$ degrees of freedom, the unconditional kurtosis of ε_t is:

$$\kappa_4 := \frac{\mathbb{E}[\varepsilon_t^4]}{(\mathbb{E}[\varepsilon_t^2])^2} = \frac{3(\nu-2)}{\nu-4} \cdot \frac{1-(\alpha+\beta)^2}{1-(\alpha+\beta)^2 - 2\alpha^2 \cdot \frac{\nu-2}{\nu-4}}, \tag{20}$$

provided $1-(\alpha+\beta)^2 - 2\alpha^2(\nu-2)/(\nu-4) > 0$. This expression is used in the moment-matching fallback estimator (activated only when MLE fails to converge) to infer α, β, and ν from the sample kurtosis of the residual series.

"The goal is not merely to model risk—it is to make sophisticated risk tools accessible to those who need them most."

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